AL-2002 - Artificial Intelligence Lab
Lab # 1

Objectives:

- Python Basics + Data Structures

Note: Carefully read the following instructions (*Each instruction contains a weightage*)

1. First think about statement problems and then write your program.
2. Write Program in multiple cells of the same notebook**.**
3. **Must Use Functions and Exception handling for each question, where they can be applied.**
4. Do not copy from any source otherwise you will be penalized with negative marks.
5. Add your source code in word document and attach output screenshots.
6. Upload word file and ipynb file.
7. Please submit your **Both files** with this naming convention ROLLNO_SECTION_LABNO. **(Do not upload zip file).**
8. Submit your lab on Google Classroom.

Question 1:

Write a program that asks the user to enter the number of students and the number of subjects. For each student, the program should accept the student's name and marks for all subjects. The program must calculate the average marks for each student and assign a grade using the following rules: average ≥ 75 as Grade A, average between 60 and 74 as Grade B, average between 40 and 59 as Grade C, and average below 40 as Fail. Finally, display each student's name, average marks, grade, and list the names of students who are academically at risk (Fail). The program must work for any number of students and subjects without using hardcoded values.

Question 2:

Write a program that first asks the user to enter the number of sales transactions for a day. For each transaction, the program should accept a product ID, quantity sold, and price per unit. After processing all transactions, the program must calculate the total revenue for the day, determine which product has the highest total quantity sold, and identify any products that were entered but had a total quantity of zero. The solution should use appropriate looping and conditional logic and should not assume a fixed number of products or transactions.

Question 3:

Write a program that accepts a sequence of integers from the user until the user enters -1. The program should count how many numbers are positive, negative, and zero, and finally display these three counts. The value -1 should be used only to stop input and must not be included in the count.

Question 4:
Write a program that asks the user to enter a string. The program must check whether the string is a palindrome without using any built-in string reversal functions. Display whether the string is a palindrome or not by comparing characters using logic and loops.

Question 5:
Write a program that builds a City Travel Tracker. Ask the user to enter city names and their directly connected cities, storing this in a dictionary. The program must:

- Add new cities and their connections
- Display all cities and their connections
- Search if a direct connection exists between two cities
- Find the city with the most connections

Use functions and exception handling throughout.